AFRICAN LEADERSHIP UNIVERSITY

[PROGRAMME]

[Software Engineering] FORMATIVE ASSIGNMENT:
[Software Development Project Proposal]

PLD ASSIGNMENT

SmartLearn

Group Members:	Contributions:
Henriette Biziyaremye	Activity Diagram
Peter Michael Angelo Rucakibungo	Use Case Diagram
Sahra Omar	Class Diagram
Keyla Nyacyesa	Flowchart Diagram
Aime Ndayambaje	Component diagram

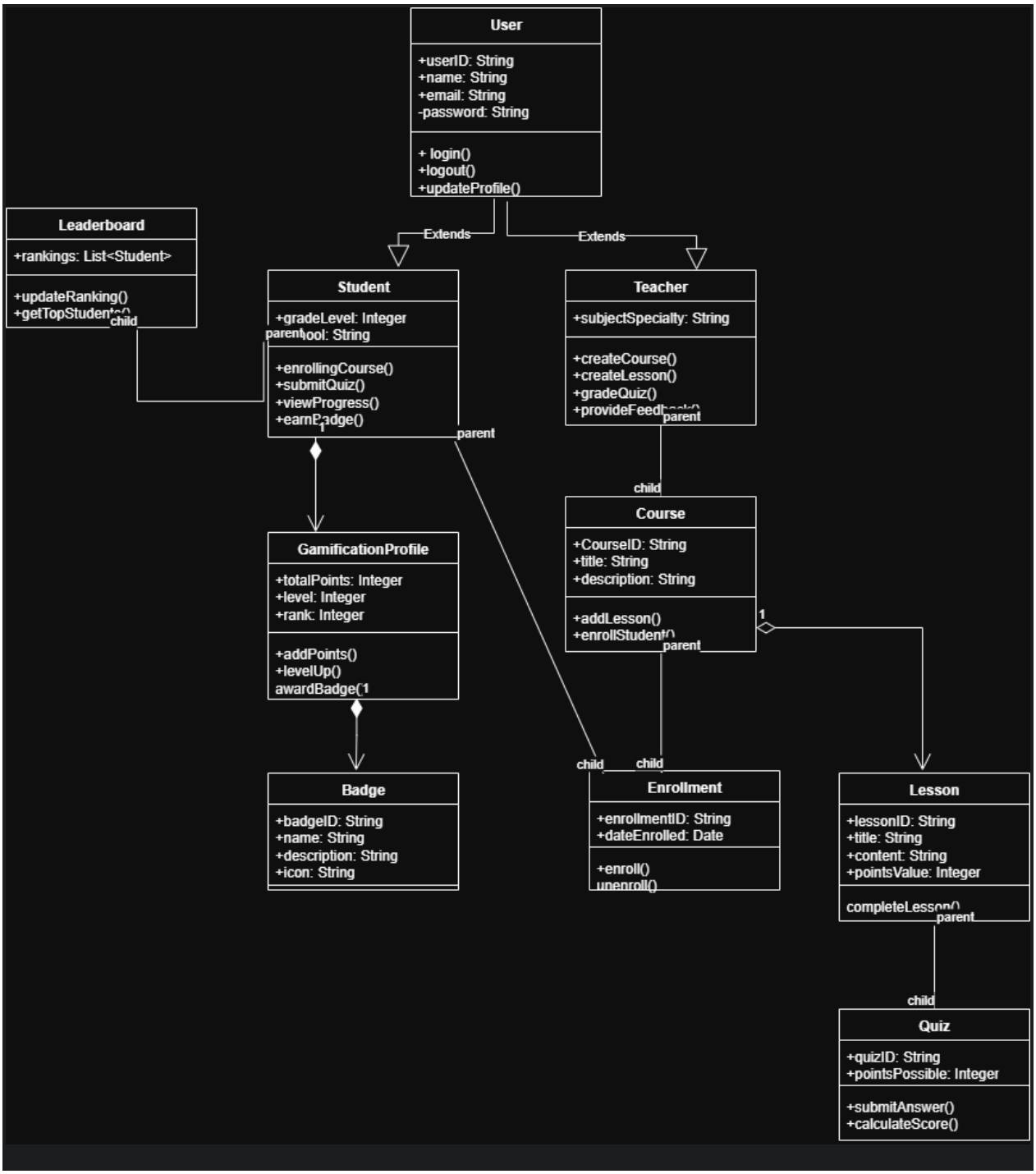
- **Description of the concept of their mission:**
  Our mission is to improve the quality of education in high school by changing the learning experience through the application of different learning delivery & experience and a change in student-teacher dynamics to foster a better environment and overall experience for the students.

- **Problem statement: A little background and identify the problem**
  Students are experiencing a lack of engagement and information retention from how the education system is structured, combined with ineffective student-teacher dynamics; this, in turn, affects them after graduation in finding employment and pursuing further education

- **The hypothesis of their solution**
  I anticipate that there will be an increase in engagement and participation once the project has gained traction in a period of 5 years and is already in use, and due to the use of gamified lessons and continuous feedback loops, students are expected to show improvement in the motivation department and a clear understanding of how their academics translate to real-world jobs

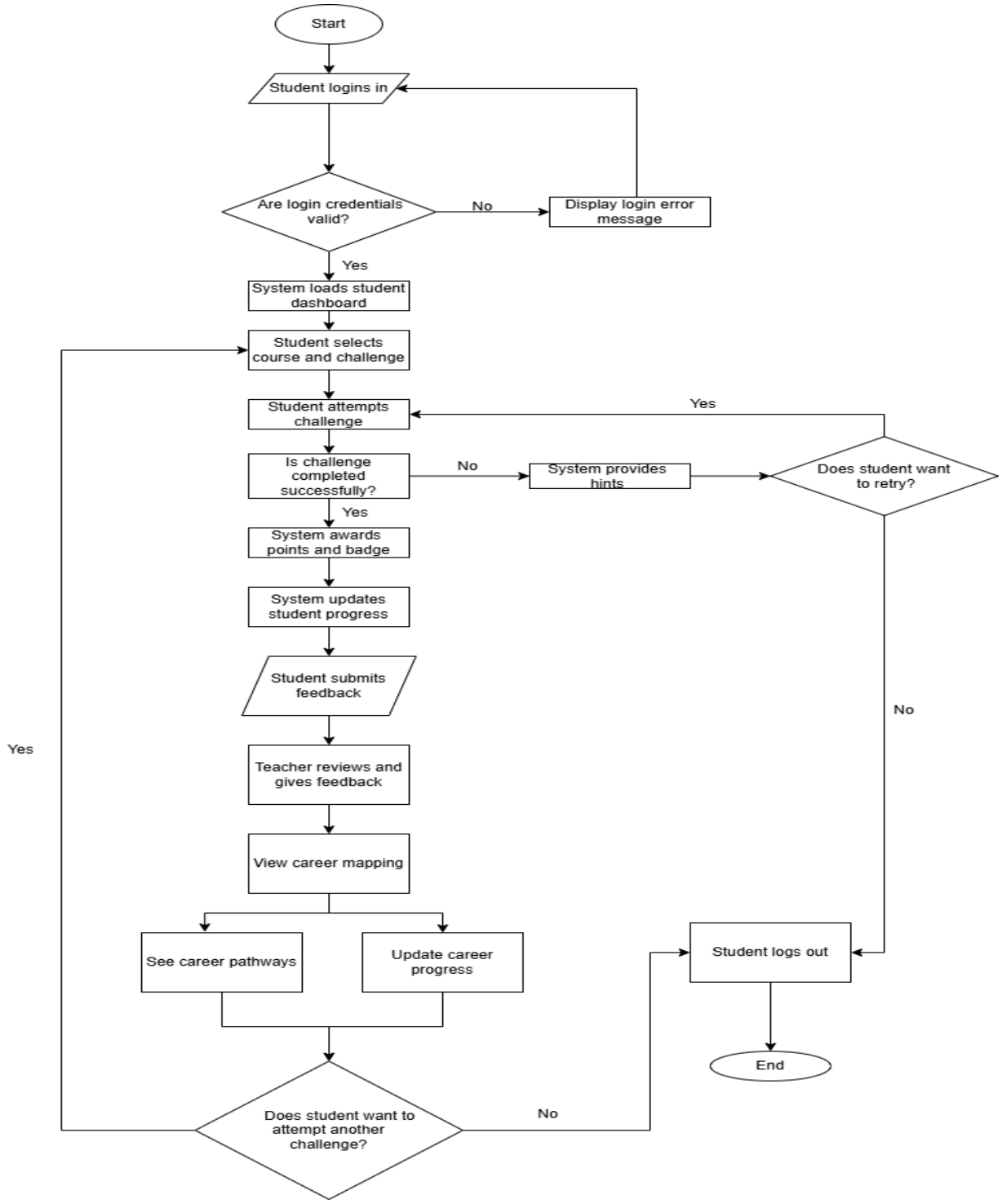
Appendix A

Class Diagram:

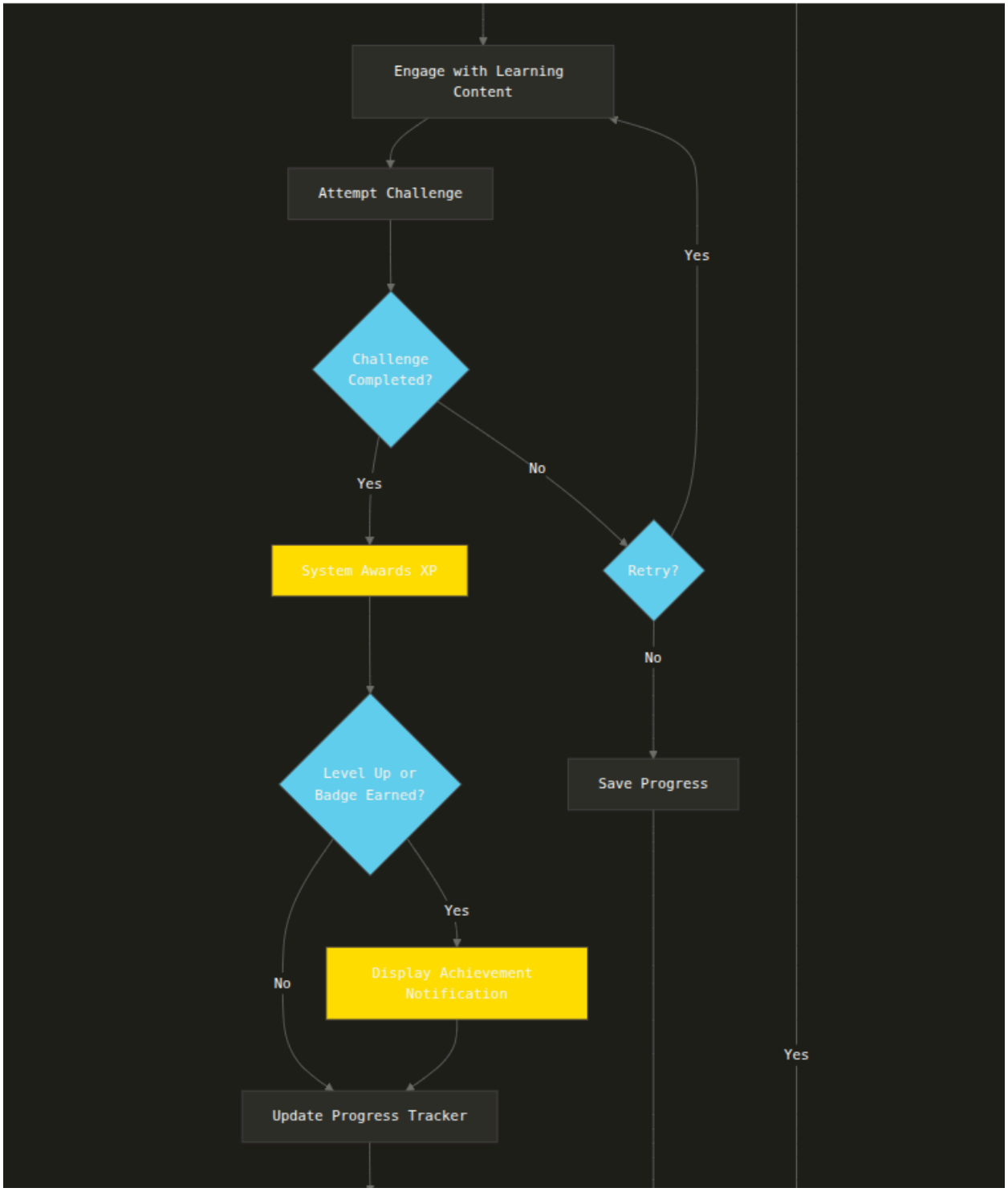


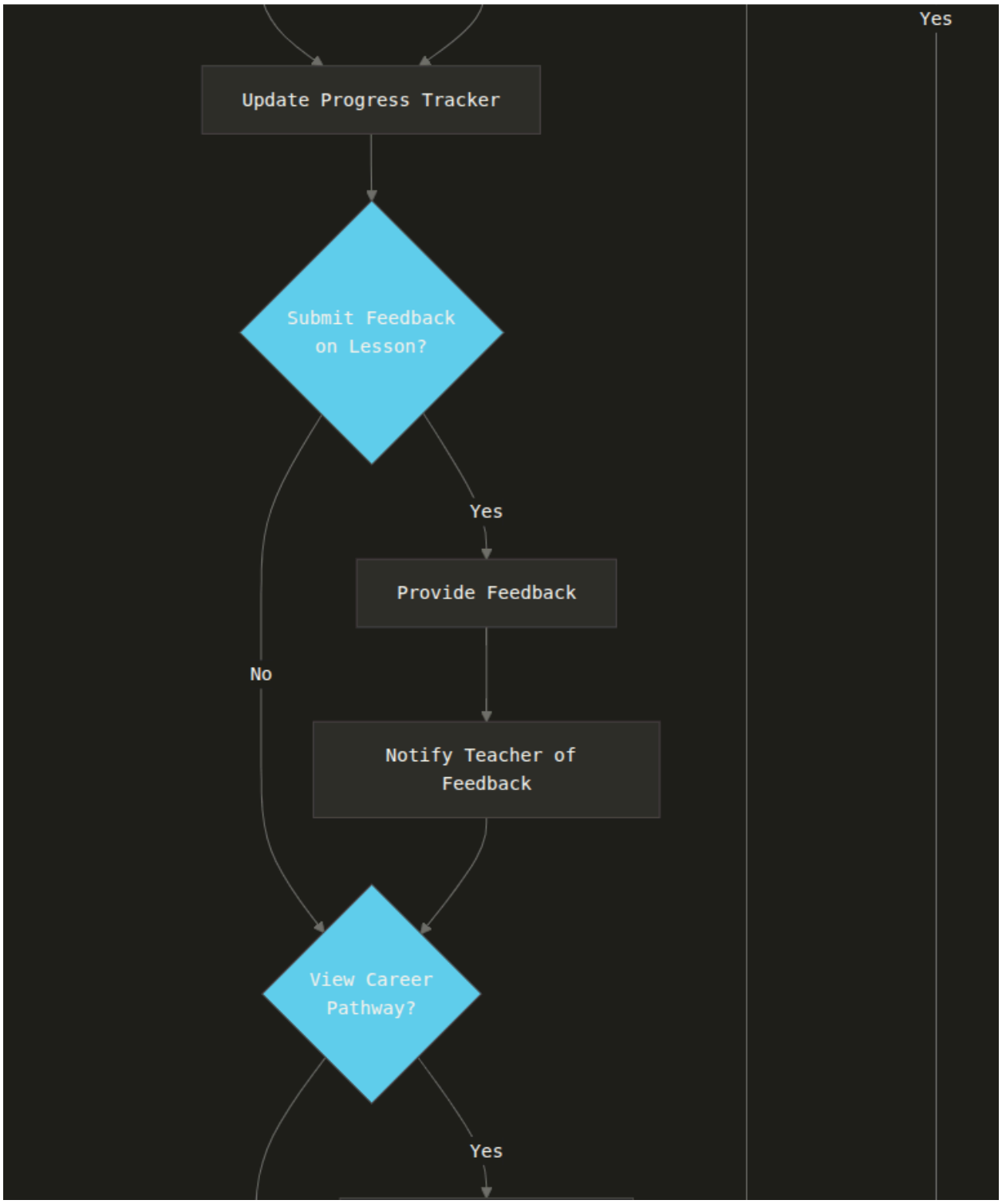
Component diagram

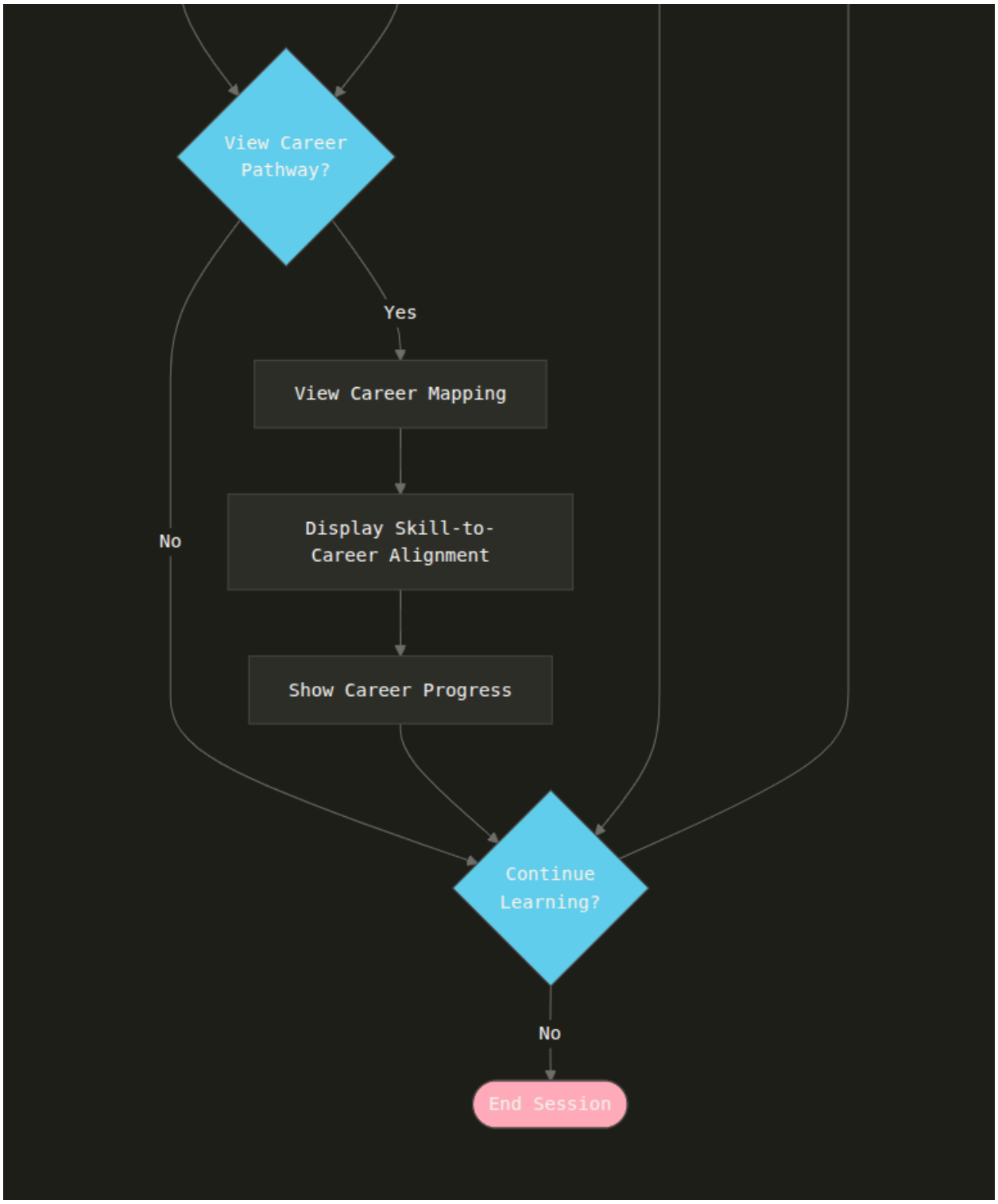
FLOWCHART:


Start
Student logins in
Are login credentials valid?
No
Display login error message
Yes
System loads student dashboard
Student selects course and challenge
Student attempts challenge
Yes
Is challenge completed successfully?
No
System provides hints
Does student want to retry?
Yes
System awards points and badge
System updates student progress
Student submits feedback
No
Yes
Teacher reviews and gives feedback
View career mapping
See career pathways
Update career progress
Student logs out
End
Does student want to attempt another challenge?
No

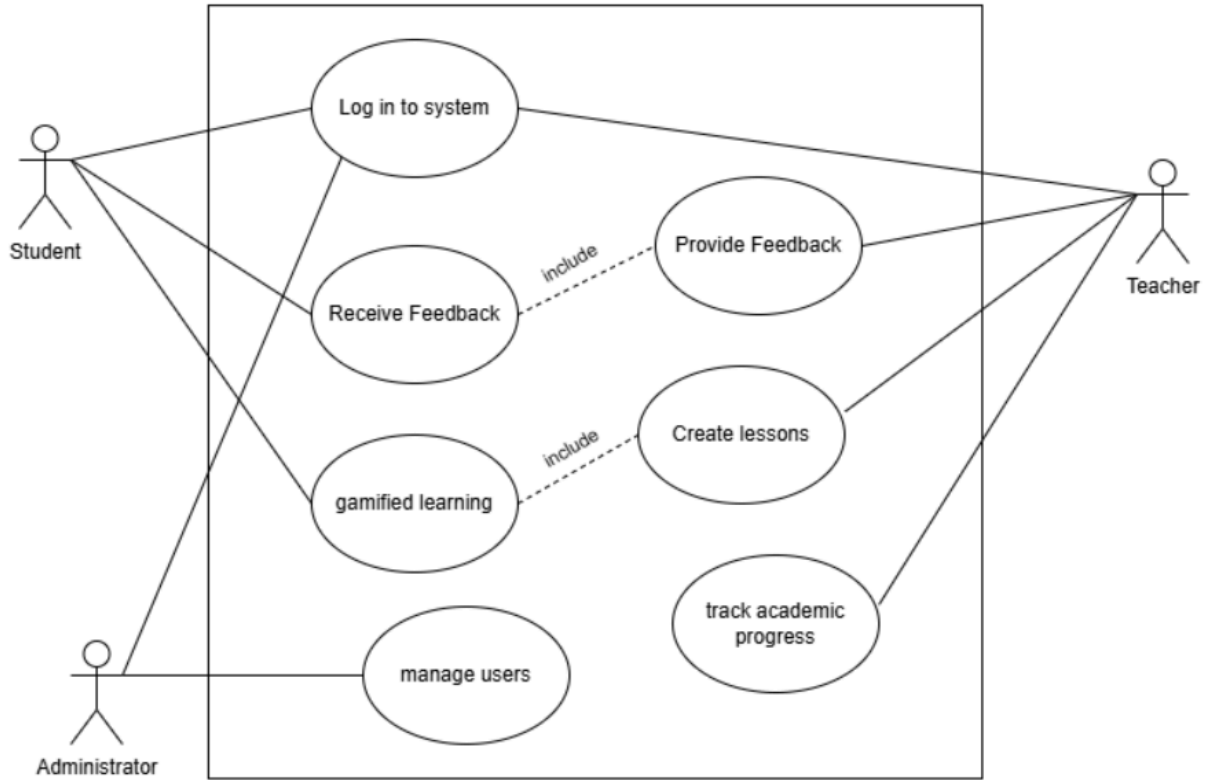
Activity Diagram:




Yes
Update Progress Tracker
Submit Feedback on Lesson?
Yes
Provide Feedback
No
Notify Teacher of Feedback
View Career Pathway?
Yes


View Career Pathway?
Yes
View Career Mapping
No
Display Skill-to-Career Alignment
Show Career Progress
Continue Learning?
No
End Session

USE CASE DIAGRAM:



(SLIDES:https://www.canva.com/design/DAHAjCDE1lk/has165HTbvGEunkFLUIUsA/edit)